

How the multi-site V1 recordings and Allen V1 differ

What we know, what protocol matching explains, and what remains unresolved

Analysis snapshot: 9 August 2026

This report summarizes the differences found so far between the eight-session multi-site V1 dataset and the Allen Visual Coding V1 data used in the original platform-paper analysis.

The recordings were collected under the project title *MouseV2*. Because that name can be mistaken for either visual area V2 or the higher visual areas, this report uses **multi-site V1 recordings** for the experimental group and **multi-site V1** as its short label. *MouseV2* refers only to the originating project, not to a brain area or dataset cohort.

The language is intentionally simple. Technical names are included only when they identify an exact analysis field or a reproducible test.

One-sentence conclusion

The multi-site V1 recordings and Allen both measure V1, but they do not yet provide a fully like-for-like absolute comparison: stimulus timing explains a large part of the grating difference and flash count explains part of the decay-time difference, while anatomical population matching and physical display timing remain unfinished.

Executive summary

The most important distinctions

1. **The anatomical question is different.** Multi-site V1 repeatedly samples four known locations within V1 in the same session. The Allen analysis pools V1 units under one VISp area label and one published hierarchy position. Allen is not literally a single-probe dataset; it is “single-ish” because V1 is represented as one area category rather than as four matched within-V1 locations.
2. **The grating stimuli differ substantially.** Multi-site V1 uses 1-second gratings, five spatial frequencies, 15 repeats, and a phase that continues across the stimulus schedule. Allen uses 2-second gratings, a fixed spatial frequency, and two different stimulus families: Brain Observatory and Functional Connectivity. Those Allen families also differ from each other.
3. **The published grating score is unusually sensitive to timing.** It first averages trials and then measures the strength of the expected response rhythm. Multi-site V1 trials have several source-defined starting phases at 1, 2, and 15 Hz. Averaging those phases cancels part of the response even when the individual trials are strongly modulated.
4. **We can now explain much, but not all, of the grating gap.** On a common 1-second, 15-trial condition set, multi-site V1 starts 0.186 and 0.220 log units below representative Allen Brain Observatory and Functional Connectivity sessions. Correcting only the known starting-phase component raises multi-site V1 by 0.117 log units and leaves gaps of 0.069 and 0.104. This removes about 53–63% of the original gap. All eight multi-site V1 sessions move in the predicted direction.
5. **The flash protocols also differ.** Multi-site V1 has 300 flashes; Allen has 150. Matching multi-site V1 to 150 flashes reduces its mean response-decay time from 47.53 to 45.92 ms. This explains 1.61 ms of the original 3.65-ms difference from Allen Brain Observatory, leaving about 2.04 ms.
6. **Absolute flash latency is not calibrated.** Multi-site V1 latency is measured from the timestamp stored in the NWB file. The files used here do not contain a photodiode or other record of the exact instant light reached the screen. We therefore do not interpret an absolute multi-site V1–Allen latency offset.
7. **Basic unit quality is now matched, but the biological populations are not.** Both datasets use the same waveform-quality rules. Multi-site V1 still lacks a validated all-session receptive-field significance/area filter comparable to Allen, and layer, depth, receptive-field location, and firing rate have not yet been matched with truly equivalent fields.

Current claim boundary

The comparison among multi-site V1 probes A, B, C, and E is meaningful as a categorical comparison among known V1 locations. The absolute Allen V1 point is still context, not a calibrated baseline. No mean shift or fitted dataset offset should be applied to make the curves line up.

1. What the two datasets actually represent

Feature	Multi-site V1 recordings (<i>MouseV2</i> project)	Allen V1 used here
Recording structure	Eight sessions from six-probe experiments; the current V1 analysis contains probes A, B, C, and E	Multi-probe experiments across many visual areas; V1 units are pooled under one VISp area label
Within-V1 replication	Four spatially separated V1 probe groups in every session	No matched four-location within-V1 design analogous to A/B/C/E in the released area-level comparison
Sessions in the released V1 summary	8	56 with valid grating values: 32 Brain Observatory and 24 Functional Connectivity
Units before common quality rules	20,374	99,180 in the full released unit table; 4,121 common-quality units in V1
Units after common quality rules	11,242	About 4,100 V1 units with valid grating values
Raw matched reanalysis	All eight multi-site V1 sessions	One representative raw session per Allen stimulus family for the common-window result, plus one reproduction-control session per family
Main strength	Repeated, simultaneous sampling of several known locations within V1	Many sessions and a broad, standardized survey of visual areas
Main limitation for this comparison	Only eight sessions and no Allen-equivalent RF/layer population yet	The published V1 value collapses within-V1 location and mixes two stimulus families

Why “single-ish V1 location” needs care

The Allen dataset is not a single recording and not necessarily a single probe trajectory. The simplification happens in the analysis: every V1 unit is given the same VISp area identity and the same published inter-area hierarchy score. The multi-site V1 recordings instead ask whether four repeated locations *inside* V1 differ from one another. These are related questions, but they are not identical experimental designs.

The unequal session structure also matters statistically. Four probe means from the same multi-site V1 session share the same animal, stimulus run, and brain state. Allen areas recorded in the same session are also correlated. Treating thousands of units as independent would greatly overstate certainty. Our current summaries therefore use session-level means, although the final model should preserve the matched groups within each session more explicitly.

Terminology used in this report

- **Modulation score:** how strongly firing follows the repeating grating.
- **Decay time:** how long the flash response persists, measured in milliseconds.
- **First-spike time:** the delay from the stored flash onset to the first detectable response.
- **Common quality rules:** the same three basic spike-sorting quality thresholds applied to both datasets.
- **Log units:** the grating scores are shown on a base-10 log scale. A change of +0.10 log units is about a 26% multiplicative increase.

2. Anatomy and location are represented differently

What is known for the multi-site V1 recordings

The multi-site V1 recordings are known experimentally to be in V1, and the part of V1 sampled by each probe is known. The missing quantity is **not** V1 identity. The missing quantity is a validated numerical hierarchy score for each probe location that would be comparable to the published numbers assigned to whole areas such as V1, LM, AL, PM, and AM.

The main analysis now handles this correctly:

- probes A, B, C, and E are treated as four categories within V1;
- small horizontal offsets around VISp in the hierarchy figure are display-only;
- no trend is fitted through those offsets;
- receptive-field coordinates are plotted separately from anatomical hierarchy.

What the current NWB files do and do not contain

All eight multi-site V1 NWBs have complete probe labels and probe-relative spike positions. However, their electrode `location` field is recorded as `unknown`, and they do not contain registered anterior–posterior, medial–lateral, or CCF coordinates. The known anatomical localization therefore exists outside the machine-readable fields currently versioned in this repository.

This does **not** invalidate the categorical probe comparison. It does prevent us from drawing a precise anatomical surface map until the targeting or registration record is imported with its coordinate system and uncertainty.

Receptive-field location is useful but not a substitute for anatomy

Provisional receptive-field peaks are available for all 20,374 multi-site V1 units and for 32 session-by-probe groups. They show that the four probes sample different parts of visual space, but they also expose important limitations:

- the expected B → C → A → E azimuth order is strictly present in only 3 of 8 sessions, or 5 of 8 when ties are allowed;
- the typical fraction of peaks on the edge of the tested grid is 0.46;
- there is no gaze-corrected all-session export;
- the current peaks do not provide a validated receptive-field significance value or area.

The measured positions are therefore a two-dimensional companion description, not a one-dimensional hierarchy score and not yet a replacement for Allen’s published receptive-field quality filter.

Layers and depth

Multi-site V1 has cortical depth and layer labels in the processed tables. They have not yet been matched to Allen with a common normalized-depth definition. A different mixture of layers could contribute to remaining response differences.

Status: V1 identity and categorical probe location are established. Exact registered surface coordinates, a within-V1 hierarchy score, and a homologous layer/RF population are not established.

3. The visual stimuli are not the same

Protocol comparison

Protocol feature	Multi-site V1	Allen Brain Observatory	Allen Functional Connectivity
Grating duration	1.0 s	2.0 s	2.0 s
Grating conditions	4 directions × 5 rates × 5 spatial frequencies = 100	40 nonblank conditions; one spatial frequency	8 directions at 2 Hz; one spatial frequency
Repeats	15 per condition	15 per condition	75 per condition
Spatial frequency	0.02, 0.04, 0.08, 0.16, 0.32 cycles/degree	0.04 cycles/degree	0.04 cycles/degree
Contrast in the common comparison	0.8	0.8	0.8
Blank after each grating	1.25 s	Different schedule	Different schedule
Grating starting phase	Frame-continuous; not reset at each presentation	Different schedule	Different schedule
Full-field flashes	150 bright + 150 dark = 300	75 bright + 75 dark = 150	75 bright + 75 dark = 150

Why running the same code was not enough

The published modulation score measures power at the expected grating rhythm relative to power at other rhythms. The time window determines which rhythms can be measured exactly.

- Multi-site V1 uses 1,000 samples from a 1-second response. Its frequency grid lands exactly on 1, 2, 4, 8, and 15 Hz.
- The released Allen calculation uses 1,024 samples from a 2-second response. Its “choose the next frequency bin” rule can evaluate a higher bin instead of the nearest one. For example, requested 2 Hz maps to 2.930 Hz rather than 1.953 Hz.

Thus, identical source code can still perform a different numerical measurement when the input window changes.

Earlier analysis mismatches that have now been fixed

These were pipeline differences, not biological findings:

- Multi-site V1 preferred gratings now include direction, movement rate, **and** spatial frequency; the earlier path pooled spatial frequencies.
- The 1.0-second window now uses exactly 1,000 one-millisecond count bins, including multiple spikes in a bin rather than binary “spike present” values.
- Flash decay fitting now uses the same 25 centered time bins as the released Allen calculation. The earlier multi-site V1 path used left edges and an extra bin.
- The same named unit-quality profile is applied in every figure and to both datasets.
- A previous plot-only shift that moved multi-site V1 first-spike times toward Allen was removed because it was not an independent display-timing calibration.

4. A simple view of what protocol matching changes

Known protocol differences explain part—but not all—of the absolute gap

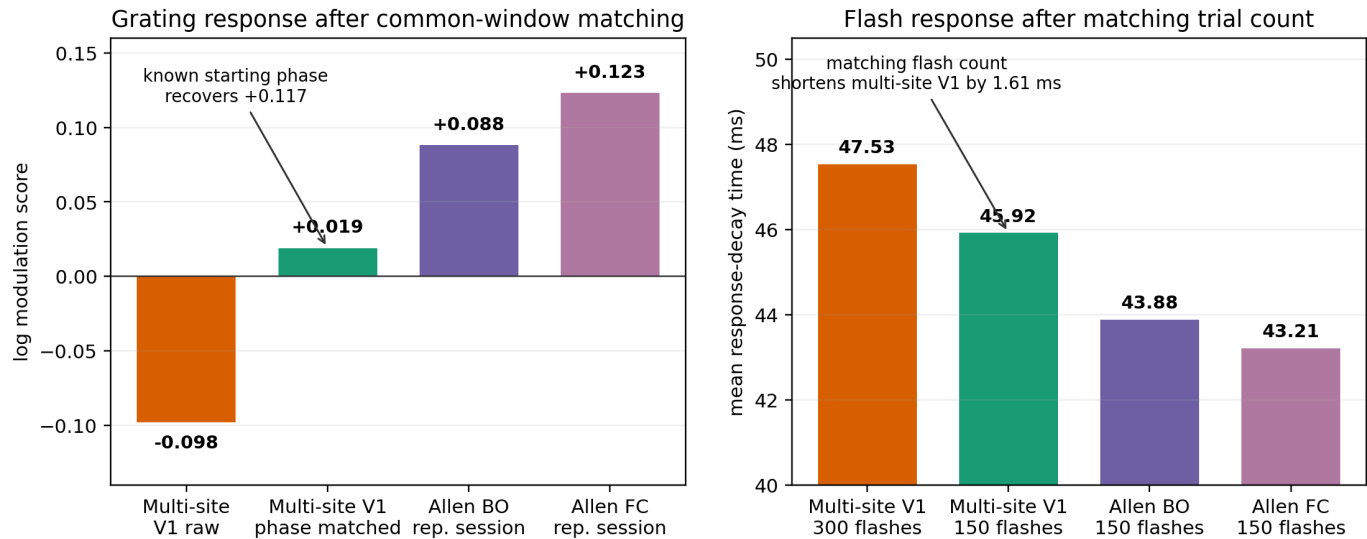


Figure 1. Left: the grating comparison after matching the response window, trial count, spatial frequency, contrast, and shared condition support. Removing only the known multi-site V1 starting-phase difference raises the multi-site V1 score but does not reach either representative Allen session. Right: matching multi-site V1 from 300 to 150 flashes shortens its estimated decay time but leaves a smaller residual difference. Allen grating bars are representative raw sessions; Allen decay bars summarize many released sessions.

What this figure does and does not say

The two panels show **protocol sensitivity**, not a fitted correction to force agreement. The multi-site V1 values change only through rules known before looking at the Allen target values: the acquisition phase schedule and the flash count.

The remaining differences can reflect several things at once:

- unmeasured protocol details;
- layer, depth, receptive-field, or firing-rate composition;
- session and animal sampling;
- genuine biological differences;
- uncertainty from using only one representative raw Allen session per stimulus family in the fully harmonized grating test.

The current evidence does not identify how much of the residual belongs to each category.

5. Grating responses: the largest discrepancy and its partial explanation

The original observation

Using the released fields with common basic quality rules, the session-average log modulation score is:

- Multi-site V1: **-0.107** across 8 sessions;
- Allen Brain Observatory: **+0.040** across 32 sessions;
- Allen Functional Connectivity: **+0.199** across 24 sessions.

The large difference between the two Allen families is itself important. It shows that there is no single protocol-independent “Allen V1 modulation value.” Functional Connectivity has many more repeats and a smaller condition set, and its published modulation score is much higher even though its alternative single-trial modulation measure is nearly unchanged.

Matching condition support did little on the multi-site V1 side

Restricting multi-site V1 to the Allen spatial frequency of 0.04 cycles/degree, four shared directions, five shared movement rates, contrast 0.8, and 15 trials changes the multi-site V1 mean by only **+0.009 log units**, from about -0.107 to -0.098. Multi-site V1's larger condition space is therefore not the main cause.

Matching the time window and trials did not remove the score gap

Representative raw Allen sessions were recomputed using the first 1 second and 15 trials. The harmonized Allen values were +0.088 for Brain Observatory and +0.123 for Functional Connectivity, compared with -0.098 for multi-site V1. The remaining raw gaps were 0.186 and 0.220 log units.

An alternative measure based on the strength within individual trials was nearly matched: multi-site V1 was only 0.032–0.035 log units below the representative Allen sessions. This told us that multi-site V1 neurons were not simply less driven by gratings.

The response is strong within trials but less aligned across trials

On the matched support:

- the mean single-trial response rhythm is comparable to or higher in multi-site V1;
- trial-to-trial phase alignment is 0.387 in multi-site V1 versus 0.539 and 0.516 in representative Allen sessions;
- Multi-site V1 loses 0.494 log units when trials are averaged, versus 0.321 and 0.345 in Allen.

In plain language, the response peaks occur at less consistent points in the grating cycle. Averaging peaks that occur at different points cancels part of the response.

The acquisition code identifies one concrete cause

Multi-site V1's grating phase is tied to the absolute frame number and does not reset at each presentation. The 1-second stimulus plus 1.25-second blank produces:

- four starting phases at 1 and 15 Hz;
- two starting phases at 2 Hz;
- one starting phase at 4 and 8 Hz.

Removing only this known starting phase raises the released-style multi-site V1 score from **−0.098 to +0.019**. All 8 sessions increase; the exact two-sided session sign test is 0.0078. Randomly permuting the phase labels gives −0.123, the wrong rotation sign gives −0.087, and the predicted 4/8-Hz controls are unchanged.

The correction raises target-rhythm power by 0.136 log units while changing the score's background spread by only 0.016. This is the expected signature of recovering a cancelled response rhythm, rather than changing the whole response.

The remaining gaps are **0.069** and **0.104 log units**. The source phase is a real and material cause, but not a complete explanation.

6. Detailed grating diagnostic

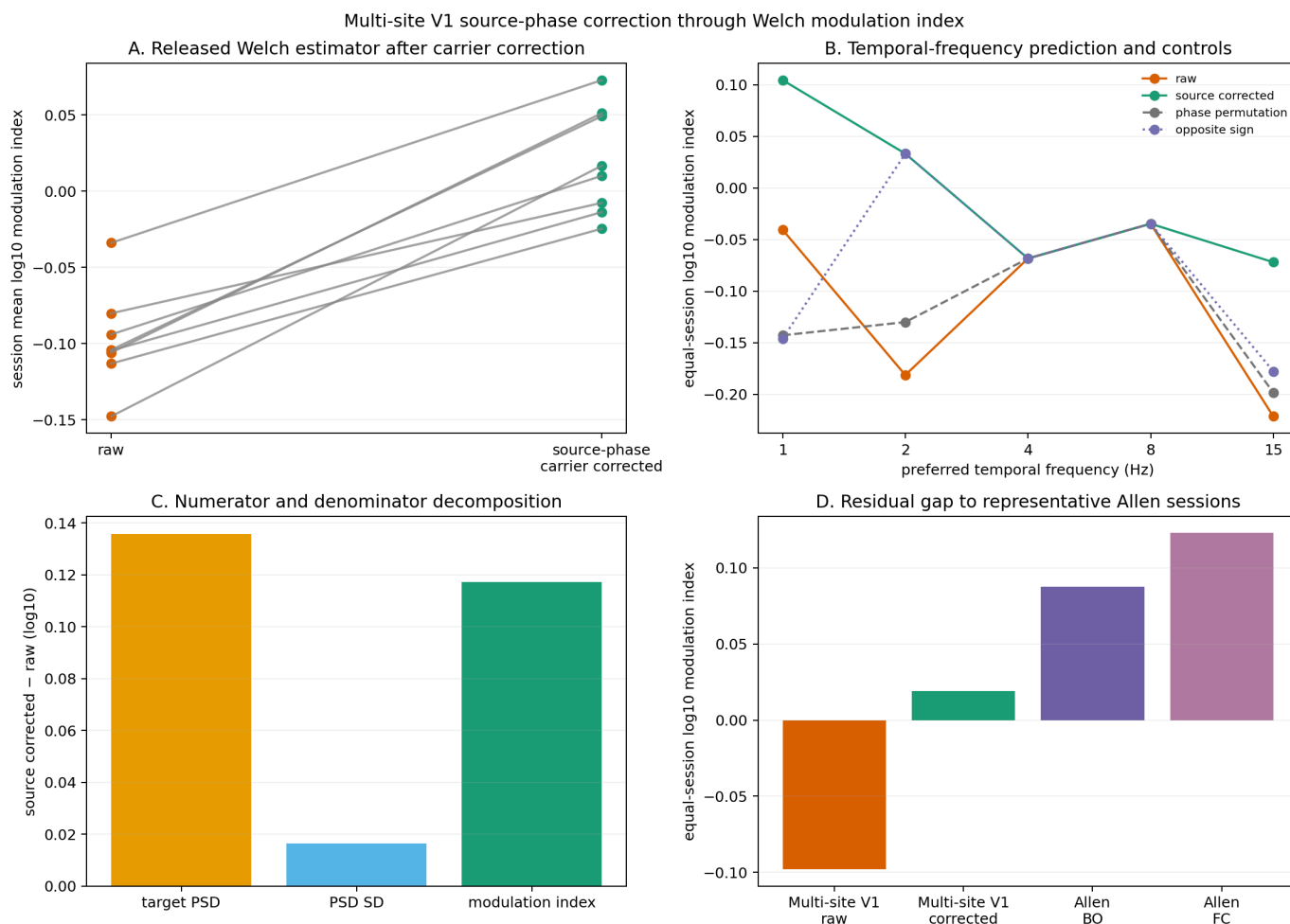


Figure 2. Panel A shows that every multi-site V1 session moves upward after the source-defined carrier correction. Panel B shows that the change occurs at the movement rates for which the acquisition schedule predicts multiple starting phases; 4 and 8 Hz do not move. Panel C shows that most of the score change comes from stronger power at the target rhythm, not from a general change in the spectrum. Panel D shows the residual gap to one representative raw Allen session from each stimulus family.

A remaining clue within the multi-site V1 recordings

After removing the known stimulus phase, a small residual phase shift is shared across physically separate probes: alignment is 0.173 versus 0.141 after shuffling trial correspondence. It also covaries descriptively with running and horizontal/vertical pupil position, but not clearly with pupil area.

This shared state does **not** close the Allen gap. Correcting a unit from the other probes changes coherence from 0.433 to 0.429 rather than increasing it. The behavior results come from only eight sessions and should be treated as clues about state, not as proof of a cause.

7. Flash responses: timing, polarity, and trial count

First-spike time

Multi-site V1 contains 150 bright and 150 dark flashes. Among common-quality units, the median first-spike time is 87.0 ms for pooled flashes, 87.0 ms for bright flashes, and 84.5 ms for dark flashes. For units with both values, the typical bright-minus-dark difference is only +1 ms.

The larger issue is the time reference. Multi-site V1 uses the NWB interval `start_time`. That timestamp matches the processed stimulus timestamp series, but the NWB does not contain an independent photodiode or physical light-onset record. A display delay could therefore shift every multi-site V1 latency without appearing in the analysis table.

For this reason:

- raw multi-site V1 latency is shown without a fitted offset;
- the former plot-only shift toward Allen has been removed;
- absolute multi-site V1–Allen first-spike differences are not interpreted;
- within-dataset differences among probes remain usable because all four probes share the same time reference within a session.

Response decay time

With all 300 multi-site V1 flashes, the equal-session mean decay time is **47.53 ms**. Allen Brain Observatory is **43.88 ms** and Functional Connectivity is **43.21 ms**, producing raw differences of +3.65 and +4.32 ms.

When multi-site V1 is repeatedly reduced to the Allen count of 75 bright plus 75 dark flashes:

- Multi-site V1 becomes **45.92 ms**;
- the change is **−1.61 ms**;
- the remaining differences are **+2.04 ms** versus Brain Observatory and **+2.71 ms** versus Functional Connectivity.

Thus trial count explains about 44% of the original Brain Observatory decay difference. It does not explain all of it.

Trial count changes which units obtain a valid decay fit

The decay calculation keeps only fits between 1 and 300 ms, with enough flash spikes and a small fitting error. With 300 flashes, 2,375 of 11,242 multi-site V1 common-quality units pass. After matching to 150 flashes, the mean valid fraction falls from 0.207 to 0.168. This selection change is part of the protocol effect, not a nuisance that can be ignored.

Bright and dark flashes are not interchangeable for decay

Bright-only and dark-only median valid decay times are about 37.26 and 37.08 ms, while the pooled median is 44.07 ms. Pooling the two polarities changes the shape and eligibility of the fitted response more than it changes first-spike time. The primary comparison therefore keeps balanced bright and dark flashes, with polarity-specific results as sensitivity checks.

8. The broader V1 comparison

V1 cross-dataset bridge: absolute metric levels are not yet calibrated

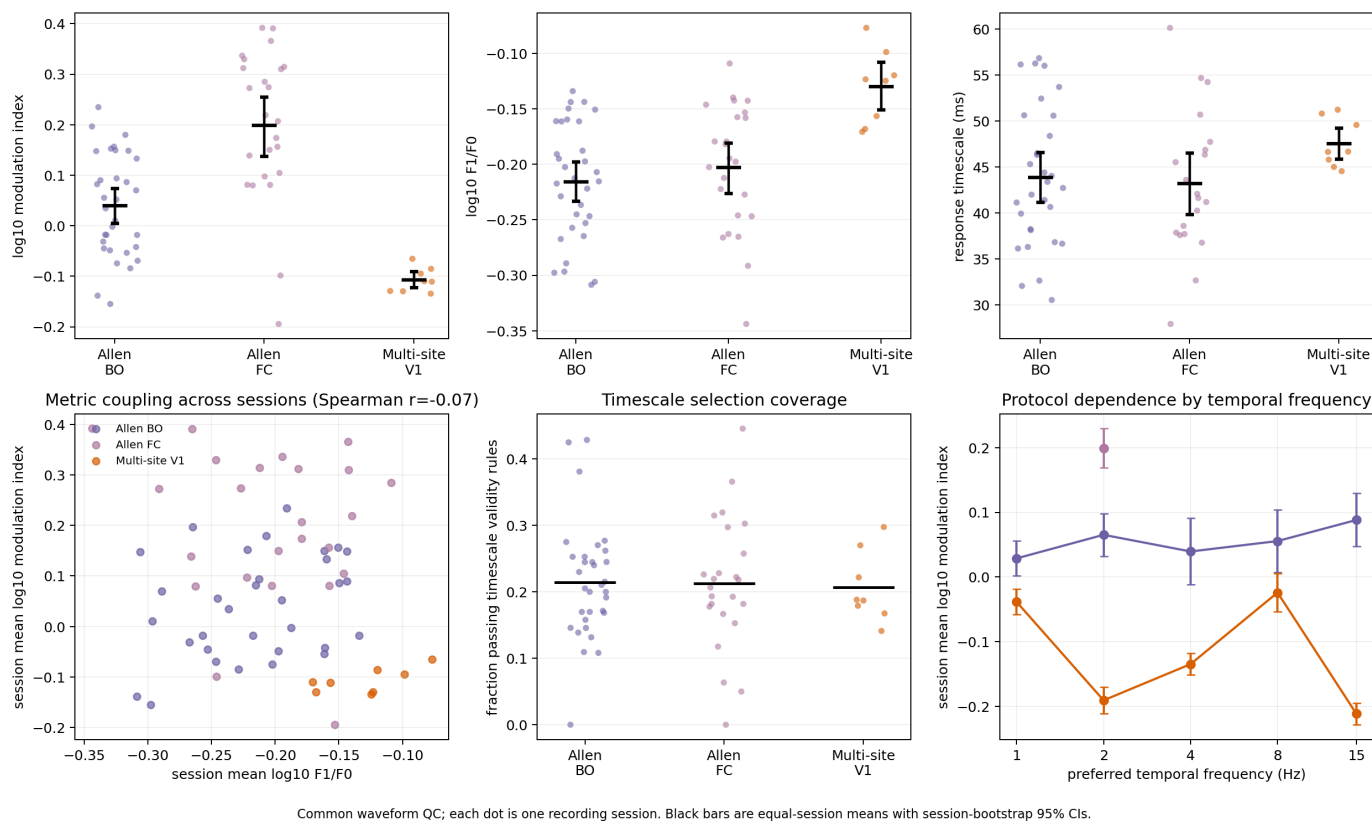


Figure 3. Each dot is one session. The upper row shows the released grating modulation score, an alternative within-trial grating measure, and flash decay time. The lower row shows relationships and selection effects. This figure is useful for seeing session-to-session spread, but its absolute centers mix the known protocol differences described in this report. The separate harmonized tests provide the fairer mechanism checks.

How to read the apparent contradictions

The first two panels can appear contradictory: multi-site V1 has a much lower published modulation score but a higher alternative grating measure. They are answering different questions. The published score rewards a response that occurs at the same phase on every trial; the alternative measure retains strong modulation within each trial even when its phase varies.

The Allen Functional Connectivity sessions also sit above Brain Observatory for the published score but not for the alternative measure. This reinforces the conclusion that the score is sensitive to trial structure and averaging, not only to the strength of the neuronal response.

The decay-time panel shows a smaller absolute difference with broad overlap between session distributions. Matching flash count narrows this difference further, but does not settle whether the residual is protocol, population, or biology.

9. Population and analysis differences that remain

Already made comparable

Item	Current status
Basic unit quality	Same thresholds in both datasets: amplitude cutoff < 0.1 , presence ratio > 0.8 , and spike-timing violation ratio < 0.5
Multi-site V1 grating preference	Uses the full direction $\times$ movement-rate $\times$ spatial-frequency condition
Common grating sensitivity	Shared directions/rates, spatial frequency 0.04, contrast 0.8, first 1 s, and 15 trials
Allen raw implementation	Reproduced released F1/F0 and modulation values in four checksum-verified NWBs
Flash polarity sensitivity	Bright, dark, and balanced pooled results are versioned
Flash trial-count sensitivity	Multi-site V1 repeatedly subsampled from 300 to 150 flashes
Multi-site V1 start phase	Reconstructed from frozen acquisition source and presentation order
Plot anatomy	Probe positions are categorical/display-only, not invented hierarchy scores

Not yet made comparable

Item	Why it matters
Multi-session raw Allen grating bridge	The fully harmonized Allen result currently uses one representative session per stimulus family; it is a diagnostic, not a population coefficient
Receptive-field significance and area	Allen can apply the published RF-quality population; multi-site V1 has provisional peaks but no validated all-session equivalent
Receptive-field position/eccentricity	Different visual-field sampling could change response composition
Layer and normalized depth	Different layer mixtures could shift modulation and decay values
Firing rate	Multi-site V1's available grating rate is for the preferred condition; Allen's released field is an overall block rate, so they cannot be used as the same covariate
Physical display onset	Multi-site V1 NWBs lack the photodiode/light-onset provenance needed for an absolute latency calibration
Registered anatomical coordinates	Known probe targeting is not yet versioned as CCF or surface coordinates in this repository
Session-matched inference	The final statistical model should preserve four probes within multi-site V1 sessions and correlated areas within Allen sessions
Statistical power	Multi-site V1 has eight sessions; its probe-effect intervals are wide compared with the larger Allen survey

Unit counts are not the number of independent experiments

Common quality rules retain 11,242 of 20,374 multi-site V1 units and about 4,100 Allen V1 units. These counts describe within-session distributions, not independent experiments. In the phase test, 4,739 affected units increase and 3,222 decrease, while every session-average increases; the session result is the primary evidence.

10. What can be said now

Findings supported by the present analysis

- Multi-site V1 samples reproducibly distinct, known locations within V1; their exact anatomical hierarchy scores are not known and are not needed for the categorical probe comparison.
- Within-V1 variation can be as large as variation among post-V1 area labels in the current three metrics. The present eight-session analysis does not show that area identity explains more variance than V1 probe identity.
- The large absolute grating-score difference is metric-specific. Multi-site V1 has strong within-trial modulation but less alignment across trials.
- The multi-site V1 acquisition phase schedule is a verified, protocol-specific cause of much of this loss. Correcting it removes roughly half to two-thirds of the harmonized representative Allen gap.
- Flash trial count explains a meaningful part of the decay-time difference and changes the set of units with valid fits.
- Bright/dark polarity matters more for decay fitting than for typical first-spike time.

Statements that are not yet supported

- That neurons in the multi-site V1 recordings are globally less modulated than Allen V1 neurons.
- That the residual corrected grating gap is biological.
- That multi-site V1 has a true absolute first-spike delay relative to Allen.
- That four multi-site V1 probes lie at four known positions on the published inter-area hierarchy axis.
- That the current populations are matched for receptive field, layer, depth, rate, or cell-type composition.
- That the two datasets are statistically equivalent.

The fairest present wording

Across eight multi-site V1 recordings, response properties vary among known locations within V1. This within-V1 variation can be comparable to the variation attributed to higher visual-area labels in the Allen dataset. Absolute multi-site V1–Allen centers are not yet calibrated: known differences in grating starting phase and flash count explain substantial portions of the modulation and decay-time offsets, while a residual remains pending broader Allen reanalysis and homologous population matching.

Practical next steps and repository ownership

1. **Current paper repository:** extend the raw Allen match, add RF/layer/rate population views, and use a session-blocked model.
2. **PilotAnalysis:** validate all-session RF significance/area, gaze correction, and the anatomical targeting export.
3. **Acquisition repositories:** preserve stimulus definitions and locate any photodiode/sync record needed for physical light-onset calibration.

Evidence and reproducibility notes

This report is a synthesis of local, versioned analyses. It does not introduce a new dataset-level correction. The main evidence files are:

- `artifacts/figure3/06b_v1_dataset_bridge/V1_DATASET_BRIDGE.md`
- `data/imports/allen_v1_raw_bridge_v2/README.md`
- `data/imports/mousev2_grating_common_support_v1/README.md`
- `data/imports/v1_grating_phase_bridge_v1/README.md`
- `data/imports/mousev2_grating_start_phase_bridge_v1/README.md`
- `data/imports/mousev2_grating_corrected_welch_bridge_v1/README.md`
- `data/imports/mousev2_grating_shared_phase_behavior_v1/README.md`
- `data/imports/mousev2_timescale_trial_bridge_v1/README.md`
- `data/imports/mousev2_flash_metrics_v1/README.md`
- `data/imports/WITHIN_V1_LOCATION_AUDIT.md`
- `artifacts/figure3/06a_known_v1_locations/Figure3_stats.md`
- `artifacts/figure3/06a_known_v1_locations/rf_position_report.md`
- `config/mousev2_stimulus_manifest.json`

Numerical validation behind the newest grating result

- 11,242 common-quality units from the multi-site V1 recordings across eight sessions were reprocessed.
- Recomputed raw modulation values match the prior bridge within 3.6×10^{-15} .
- Reconstructing the raw response from carrier and non-carrier components has a maximum error of 5.6×10^{-17} .
- Mean firing rate changes by at most 4.2×10^{-17} during the target-only phase correction.
- All 44 repository regression tests pass.

Final interpretation rule

When a difference disappears after a pre-specified protocol match, it should not be called biological. When a difference remains, it is a candidate for biology only after timing, anatomical population, and sampling differences have been addressed.

Current overall status: improved comparability, but not yet a fair absolute calibration of the Allen V1 point.